

EMPLOYEE

```
1 CREATE TABLE EMPLOYEE (  
2     SSN VARCHAR2(20) PRIMARY KEY,  
3     BirthDate DATE,  
4     Gender VARCHAR2(10),  
5     Fname VARCHAR2(50),  
6     Lname VARCHAR2(50),  
7     Supervisor VARCHAR2(20),  
8     FOREIGN KEY (Supervisor) REFERENCES Employee(SSN)  
9 );
```

DEPARTMENT

```
1 CREATE TABLE Department (  
2     DNUM NUMBER PRIMARY KEY,  
3     DName VARCHAR2(100),  
4     SSN VARCHAR2(20),  
5     HiringDate DATE,  
6     FOREIGN KEY (SSN) REFERENCES Employee(SSN)  
7 );
```

PROJECT

```
1 CREATE TABLE Project (  
2     PNumber NUMBER PRIMARY KEY,  
3     PName VARCHAR2(100),  
4     Location VARCHAR2(100),  
5     City VARCHAR2(100),  
6     DNUM NUMBER,  
7     FOREIGN KEY (DNUM) REFERENCES Department(DNUM)  
8 );
```

DEPENDENT

```
1 CREATE TABLE Dependent (  
2     DependentName VARCHAR2(100),  
3     BirthDate DATE,  
4     Gender VARCHAR2(10),  
5     SSN VARCHAR2(20),  
6     FOREIGN KEY (SSN) REFERENCES Employee(SSN)  
7 );
```

WORKS_ON

```
1 CREATE TABLE Works_On (  
2     SSN VARCHAR2(20),  
3     PNumber NUMBER,  
4     WorkingHours NUMBER,  
5     FOREIGN KEY (SSN) REFERENCES Employee(SSN),  
6     FOREIGN KEY (PNumber) REFERENCES Project(PNumber),  
7     PRIMARY KEY (SSN, PNumber)  
8 );
```

LOCATIONS

```
1 CREATE TABLE Locations (  
2     Location VARCHAR2(100),  
3     DNUM NUMBER,  
4     FOREIGN KEY (DNUM) REFERENCES Department(DNUM),  
5     PRIMARY KEY (Location, DNUM)  
6 );
```

Alter:

```
1 ALTER TABLE EMPLOYEE  
2 MODIFY Gender CHAR(1);  
3  
4 ALTER TABLE Dependent  
5 MODIFY Gender CHAR(1);  
6  
7 ALTER TABLE EMPLOYEE  
8 ADD DNUM NUMBER;  
9  
10 ALTER TABLE EMPLOYEE  
11 ADD CONSTRAINT fk_employee_department  
12 FOREIGN KEY (DNUM) REFERENCES Department(DNUM);
```